

1 Summary

Model (Run-name): model3 (test)

Iteration: 7

TAC: 63207.29376333453

Took: 56.95697259902954s

Total Time: 203.51903223991394s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

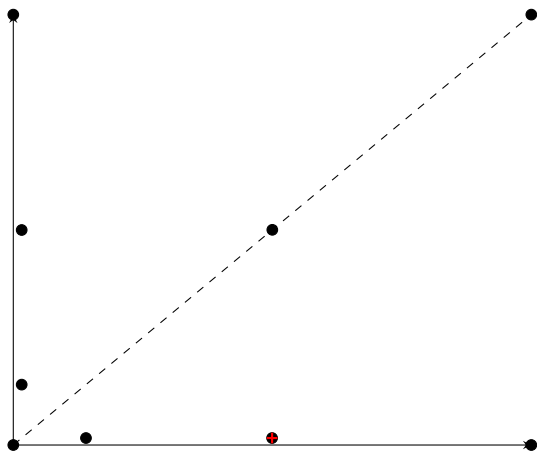
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{1,4,2}$	1.0	$dt_{3,5,1}$	22.3	$q_{1,4,2}$	586.96
$z_{2,2,1}$	1.0	$dt_{3,5,2}$	29.05608108108106	$q_{2,2,1}$	647.2099999999999
$z_{2,4,1}$	1.0	$dt_{4,3,1}$	50.0	$q_{2,4,1}$	530.45
$z_{3,5,1}$	1.0	$dt_{4,3,2}$	111.1	$q_{3,5,1}$	1544.29
$z_{4,3,1}$	1.0	$dt_{5,1,1}$	38.9	$q_{4,3,1}$	1539.7199999999998
$z_{5,1,1}$	1.0	$dt_{5,1,2}$	66.72824858757063	$q_{5,1,1}$	760.0
$z_{5,4,1}$	1.0	$dt_{5,4,1}$	22.2	$q_{5,4,1}$	517.43999999999996
$z_{cu,3}$	1.0	$dt_{5,4,2}$	10.599924888148678	$q_{cu,3}$	839.9899999999999
$z_{cu,5}$	1.0	$dt_{cu,3}$	82.35608108108107	$q_{cu,5}$	1081.9699999999998
$A_{1,4,2}^\beta$	7.4535714398973045	$dt_{cu,5}$	86.72824858757062	$q_{1,1}^{(H)}$	160.0
$A_{2,2,1}^\beta$	7.838832697204902	$f_{1,4,2}^H$	8.8	$q_{1,2}^{(H)}$	160.0
$A_{2,4,1}^\beta$	0.5207742819827617	$f_{2,2,1}^H$	5.825472547254726	$q_{1,3}^{(H)}$	93.3
$A_{3,5,1}^\beta$	12.92300279832656	$f_{2,4,1}^H$	4.774527452745275	$q_{2,1}^{(H)}$	248.9
$A_{4,3,1}^\beta$	6.6761148025763255	$f_{3,5,1}^H$	9.585909373060211	$q_{2,2}^{(H)}$	137.8
$A_{5,1,1}^\beta$	5.152662429360161	$f_{4,3,1}^H$	12.6	$q_{2,3}^{(H)}$	137.8
$A_{5,4,1}^\beta$	2.518671738671132	$f_{5,1,1}^H$	5.701425356339083	$q_{3,1}^{(H)}$	226.7
$A_{cu,3}^\beta$	5.422960329199584	$f_{5,4,1}^H$	3.881770442610649	$q_{3,2}^{(H)}$	122.35608108108107
$A_{cu,5}^\beta$	6.2036407384972705	$f_{1,4,2}^C$	12.443378249203791	$q_{3,3}^{(H)}$	122.35608108108107
$A_{1,4,2}$	28.95876233416589	$f_{2,2,1}^C$	6.1	$q_{4,1}^{(H)}$	271.1
$A_{2,2,1}$	30.93278282491164	$f_{2,4,1}^C$	8.757393428699588	$q_{4,2}^{(H)}$	148.9
$A_{2,4,1}$	3.743604220332406	$f_{3,5,1}^C$	13.9	$q_{4,3}^{(H)}$	148.9
$A_{3,5,1}$	71.16597394333036	$f_{4,3,1}^C$	8.4	$q_{5,1}^{(H)}$	198.9
$A_{4,3,1}$	23.670439786681666	$f_{5,1,1}^C$	7.6	$q_{5,2}^{(H)}$	126.72824858757063
$A_{5,1,1}$	16.623773924385056	$f_{5,4,1}^C$	8.542606571300412	$q_{5,3}^{(H)}$	126.72824858757062
$A_{5,4,1}$	7.000222767756085	$RecLMTD_{1,4,2}$	0.04193632953530227	$q_{1,1}^{(C)}$	160.0
$A_{cu,3}$	16.73892021814917	$RecLMTD_{2,2,1}$	0.040624936884743584	$q_{1,2}^{(C)}$	60.0
$A_{cu,5}$	20.94487303718885	$RecLMTD_{2,4,1}$	0.025165980106347276	$q_{1,3}^{(C)}$	60.0
$dt_{1,4,2}$	43.87167630057803	$RecLMTD_{3,5,1}$	0.039170802020236366	$q_{2,1}^{(C)}$	221.7
$dt_{1,4,3}$	11.1	$RecLMTD_{4,3,1}$	0.013067228988828773	$q_{2,2}^{(C)}$	115.6
$dt_{2,2,1}$	27.2	$RecLMTD_{5,1,1}$	0.01859237873122012	$q_{2,3}^{(C)}$	115.6
$dt_{2,2,2}$	22.2	$RecLMTD_{5,4,1}$	0.05629902886046477	$q_{3,1}^{(C)}$	221.1
$dt_{2,4,1}$	72.2	$RecLMTD_{cu,3}$	0.016938394725445276	$q_{3,2}^{(C)}$	37.8
$dt_{2,4,2}$	21.671676300578056	$RecLMTD_{cu,5}$	0.016454376814154292	$q_{3,3}^{(C)}$	37.8

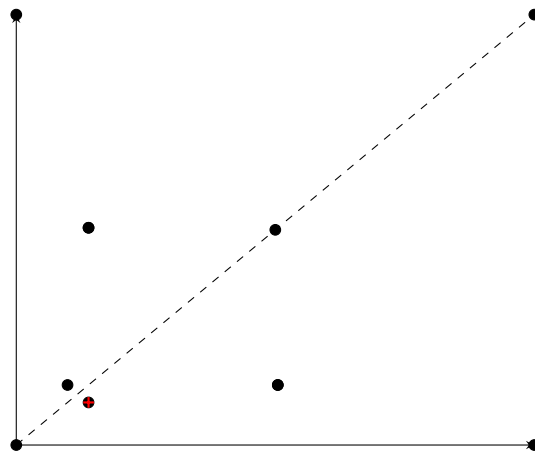
Variable	Value	Variable	Value
$t_{4,1}^{(C)}$	176.7	$b_{5,4,1}^{H,out}$	254.64414103525854
$t_{4,2}^{(C)}$	116.12832369942195	$b_{1,4,2}^{C,in}$	1022.8456920845516
$t_{4,3}^{(C)}$	82.2	$b_{2,2,1}^{C,in}$	705.16
$t_{5,1}^{(C)}$	204.4	$b_{2,4,1}^{C,in}$	1016.9814188512169
$t_{5,2}^{(C)}$	93.3	$b_{3,5,1}^{C,in}$	1296.8700000000003
$t_{5,3}^{(C)}$	93.3	$b_{4,3,1}^{C,in}$	317.5200000000002
$t_{1,4,2}^H$	93.3	$b_{5,1,1}^{C,in}$	456.0
$t_{2,2,1}^H$	137.8	$b_{5,4,1}^{C,in}$	992.0385811487831
$t_{2,4,1}^H$	198.8575471698113	$b_{1,4,2}^{C,out}$	1609.8056920845518
$t_{3,5,1}^H$	65.6	$b_{2,2,1}^{C,out}$	1352.37
$t_{4,3,1}^H$	148.9	$b_{2,4,1}^{C,out}$	1547.4314188512171
$t_{5,1,1}^H$	65.6	$b_{3,5,1}^{C,out}$	2841.1600000000003
$t_{5,4,1}^H$	65.6	$b_{4,3,1}^{C,out}$	1857.24
$t_{1,4,2}^C$	129.3704699676358	$b_{5,1,1}^{C,out}$	1216.0
$t_{2,2,1}^C$	221.7	$b_{5,4,1}^{C,out}$	1509.4785811487827
$t_{2,4,1}^C$	147.35163152253665		
$t_{3,5,1}^C$	204.4		
$t_{4,3,1}^C$	221.1		
$t_{5,1,1}^C$	160.0		
$t_{5,4,1}^C$	176.7		
$b_{1,4,2}^{H,in}$	1408.0		
$b_{2,2,1}^{H,in}$	1449.9601170117014		
$b_{2,4,1}^{H,in}$	1188.379882988299		
$b_{3,5,1}^{H,in}$	2173.1256548727497		
$b_{4,3,1}^{H,in}$	3415.8599999999999		
$b_{5,1,1}^{H,in}$	1134.0135033758438		
$b_{5,4,1}^{H,in}$	772.0841410352582		
$b_{1,4,2}^{H,out}$	821.04		
$b_{2,2,1}^{H,out}$	802.7501170117015		
$b_{2,4,1}^{H,out}$	657.9298829882986		
$b_{3,5,1}^{H,out}$	628.8356548727498		
$b_{4,3,1}^{H,out}$	1876.1399999999996		
$b_{5,1,1}^{H,out}$	374.01350337584387		

3 *RecLMTD* Tangent Points

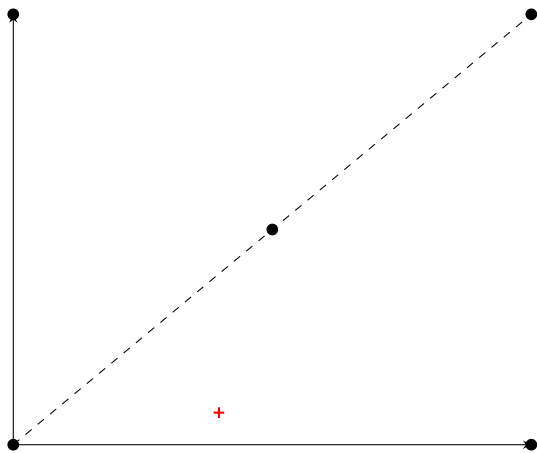
- (1, 4, 2)
 - Value: 0.041936
 - Error: 0.0000000000000000
 - Relative error: -0.0000000000000000
- (2, 2, 1)
 - Value: 0.040625
 - Error: 0.0000000000000000
 - Relative error: -0.0000000000000001
- (2, 4, 1)
 - Value: 0.025166
 - Error: 0.001348964432463
 - Relative error: -0.056638684331145
- (3, 5, 1)
 - Value: 0.039171
 - Error: 0.0000000000000000
 - Relative error: -0.0000000000000000
- (4, 3, 1)
 - Value: 0.013067
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (5, 1, 1)
 - Value: 0.018592
 - Error: 0.000799216395438
 - Relative error: 0.041214577254639
- (5, 4, 1)
 - Value: 0.056299
 - Error: 0.007428607959212
 - Relative error: 0.116568075170143
- $cu, 3$
 - Value: 0.016938
 - Error: 0.000000077787154
 - Relative error: 0.000004592335795
- $cu, 5$
 - Value: 0.016454
 - Error: 0.0000000000000000
 - Relative error: -0.0000000000000000



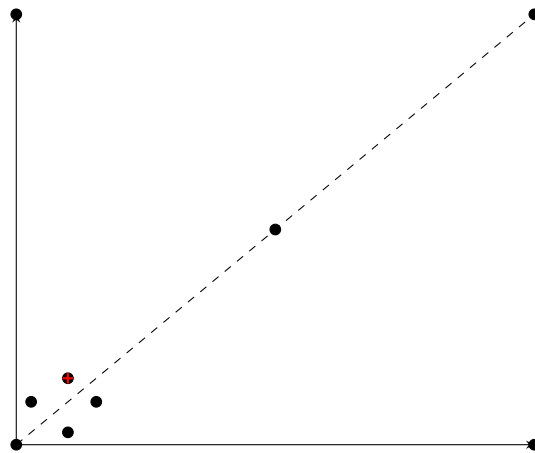
(a) RecLMTD - (1, 4, 2)



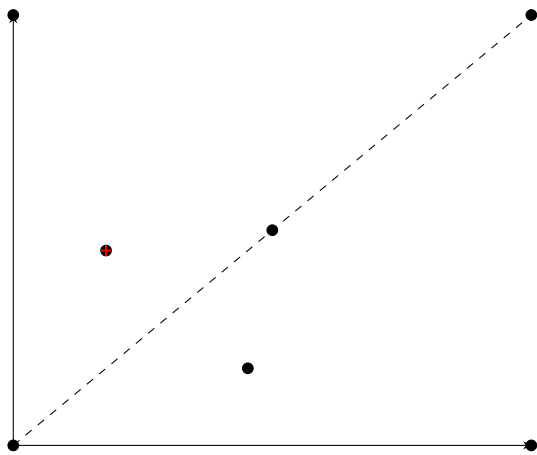
(b) RecLMTD - (2, 2, 1)



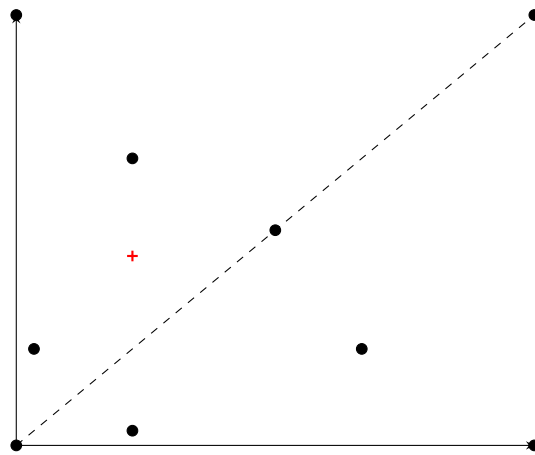
(c) RecLMTD - (2, 4, 1)



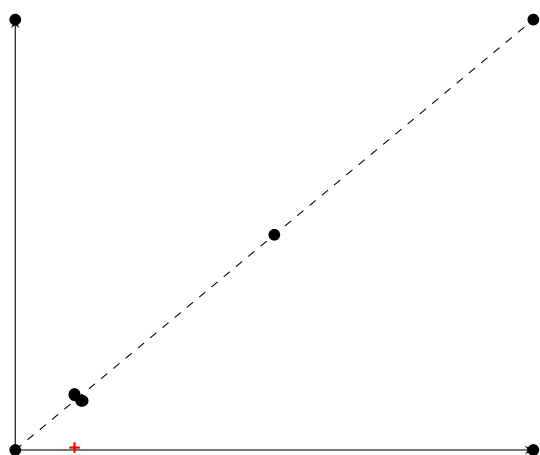
(d) RecLMTD - (3, 5, 1)



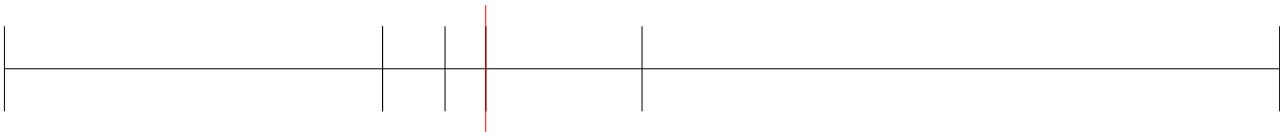
(e) RecLMTD - (4, 3, 1)



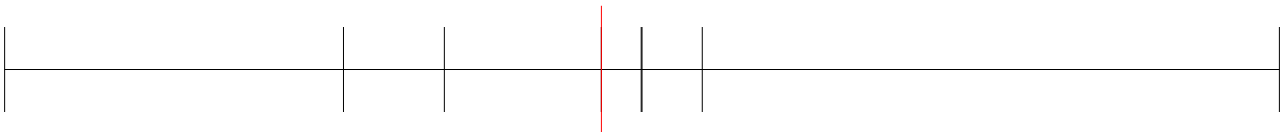
(f) RecLMTD - (5, 1, 1)



(g) *RecLMTD* - (5, 4, 1)



(a) *RecLMTD* - *cu*, 3



(b) *RecLMTD* - *cu*, 5

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 4, 2)
 - Value: 1408.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1449.9601170117014
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 4, 1)
 - Value: 1188.379882988299
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 1)
 - Value: 2173.1256548727497
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 1)
 - Value: 3415.8599999999999
 - Absolute error: 9.094947017729282e-13
 - Relative error: -2.6625643374521444e-16
- (5, 1, 1)
 - Value: 1134.0135033758438
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 2.0050349909093993e-16
- (5, 4, 1)
 - Value: 772.0841410352582
 - Absolute error: 1.1368683772161603e-13
 - Relative error: 1.4724669460141701e-16

4.2 $b^{H,\text{out}}$

- (1, 4, 2)
 - Value: 821.04
 - Absolute error: 1.1368683772161603e-13
 - Relative error: -1.3846686851020172e-16
- (2, 2, 1)

- Value: 802.7501170117015
 - Absolute error: 1.1368683772161603e-13
 - Relative error: 1.4162170183770757e-16
- (2, 4, 1)
 - Value: 657.9298829882986
 - Absolute error: 291.520935159554
 - Relative error: -0.30704163879519364
- (3, 5, 1)
 - Value: 628.8356548727498
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 1)
 - Value: 1876.1399999999996
 - Absolute error: 4.547473508864641e-13
 - Relative error: -2.4238455066597593e-16
- (5, 1, 1)
 - Value: 374.01350337584387
 - Absolute error: 5.684341886080802e-14
 - Relative error: 1.5198226360208825e-16
- (5, 4, 1)
 - Value: 254.64414103525854
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,in}$

- (1, 4, 2)
 - Value: 1022.8456920845516
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 4, 1)
 - Value: 1016.9814188512169
 - Absolute error: 3.410605131648481e-13
 - Relative error: 3.3536553062111055e-16

- (3, 5, 1)
 - Value: 1296.8700000000003
 - Absolute error: 4.547473508864641e-13
 - Relative error: 3.506499116229569e-16
- (4, 3, 1)
 - Value: 317.52000000000002
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 7.160924522651551e-16
- (5, 1, 1)
 - Value: 456.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 1)
 - Value: 992.0385811487831
 - Absolute error: 3.410605131648481e-13
 - Relative error: -3.437976300981148e-16

4.4 $b^{C,out}$

- (1, 4, 2)
 - Value: 1609.8056920845518
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 4, 1)
 - Value: 1547.4314188512171
 - Absolute error: 257.01520924759166
 - Relative error: 0.19917233473573498
- (3, 5, 1)
 - Value: 2841.1600000000003
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 1)
 - Value: 1857.24
 - Absolute error: 0.0

- Relative error: 0.0
- (5, 1, 1)
 - Value: 1216.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 1)
 - Value: 1509.4785811487827
 - Absolute error: 0.0
 - Relative error: 0.0

$$(a) \ t^{(H)} - (1, 2)$$

$$(a) \ t^{(H)} - (2, 1)$$

$$(a) \ t^{(H)} - (2, 1)$$

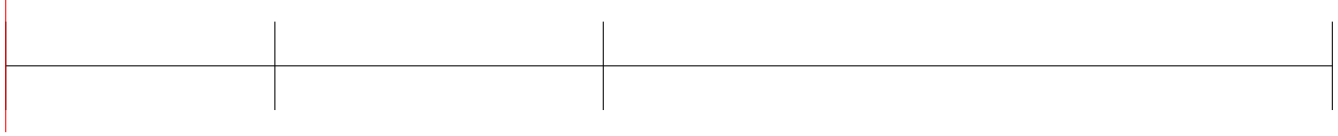
$$(a) \ t^{(H)} - (3, 1)$$

$$(a) \ t^{(H)} - (4, 1)$$

$$(a) \ t^{(H)} - (5, 1)$$

$$(a) \ t^{(H)} - (5, 1)$$

$$(a) \ t^{(H)} - (1, 4, 2)$$



(a) $t^{(H)} - (2, 2, 1)$



(a) $t^{(H)} - (2, 4, 1)$



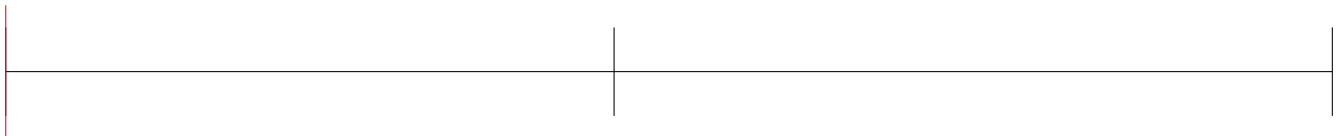
(a) $t^{(H)} - (3, 5, 1)$



(a) $t^{(H)} - (4, 3, 1)$



(a) $t^{(H)} - (5, 1, 1)$



(a) $t^{(H)} - (5, 4, 1)$



(a) $t^{(C)} - (4, 3)$



(a) $t^{(C)} - (2, 2)$



(a) $t^{(C)} - (4, 2)$



(a) $t^{(C)} - (5, 2)$



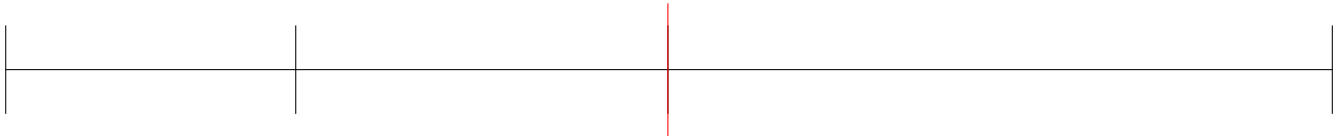
(a) $t^{(C)} - (3, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (4, 2)$



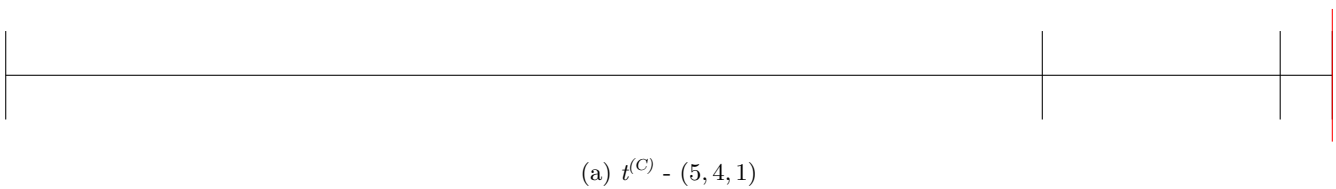
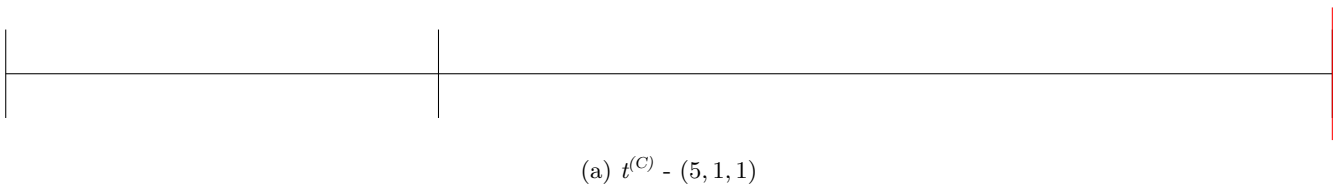
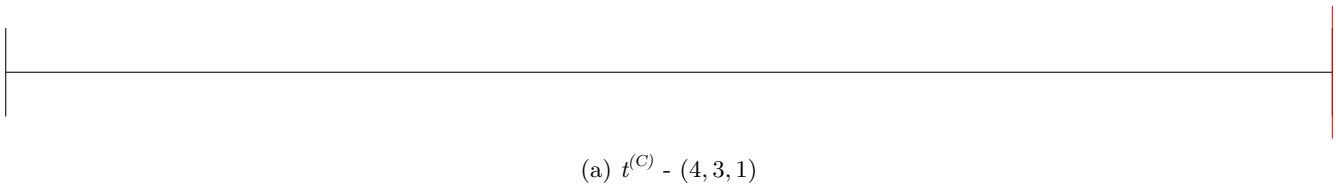
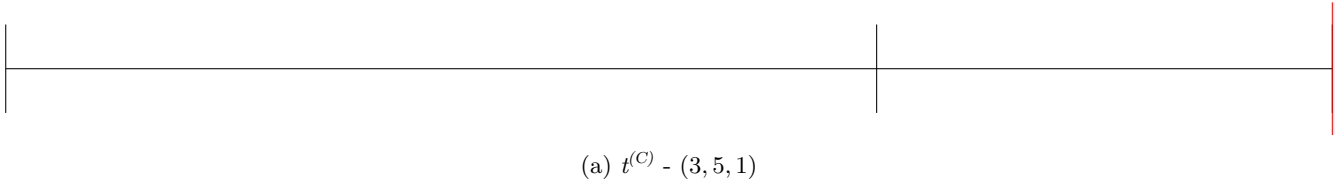
(a) $t^{(C)} - (1, 4, 2)$



(a) $t^{(C)} - (2, 2, 1)$



(a) $t^{(C)} - (2, 4, 1)$



5 Area Breakpoints

5.1 A

- $(1, 4, 2)$
 - Value: 28.958762
 - Error: 0.000000000000018
 - Relative error: 0.000000000000001
- $(2, 2, 1)$
 - Value: 30.932783
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- $(2, 4, 1)$
 - Value: 3.743604
 - Error: 11.961447717799258
 - Relative error: 0.761630573710935
- $(3, 5, 1)$
 - Value: 71.165974
 - Error: 0.000000000000014
 - Relative error: 0.000000000000000
- $(4, 3, 1)$
 - Value: 23.670440
 - Error: 0.000000000000025
 - Relative error: 0.000000000000001
- $(5, 1, 1)$
 - Value: 16.623774
 - Error: 0.000000000000007
 - Relative error: -0.000000000000000
- $(5, 4, 1)$
 - Value: 7.000223
 - Error: 27.271976636430821
 - Relative error: 0.795746322399697

5.2 A_{CU}

- $cu, 3$
 - Value: 16.738920
 - Error: 0.000000000000021
 - Relative error: -0.000000000000001
- $cu, 5$

- Value: 20.944873
- Error: 0.0000000000000004
- Relative error: -0.0000000000000000



(a) $A^\beta - (1, 4, 2)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (2, 4, 1)$



(a) $A^\beta - (3, 5, 1)$



(a) $A^\beta - (4, 3, 1)$



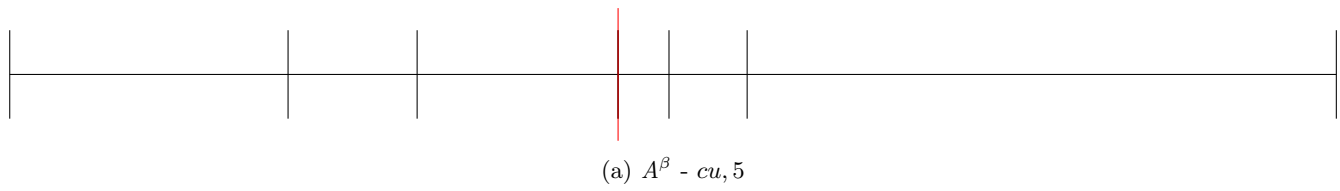
(a) $A^\beta - (5, 1, 1)$



(a) $A^\beta - (5, 4, 1)$



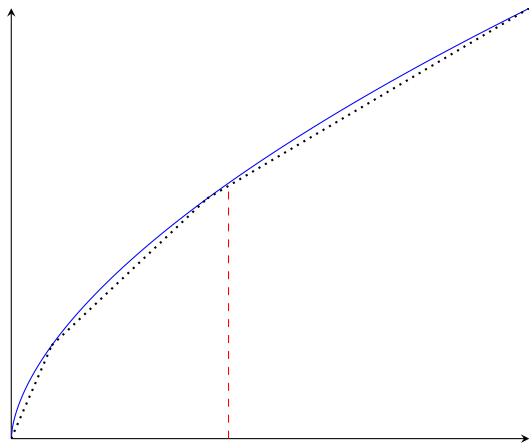
(a) $A^\beta - cu, 3$



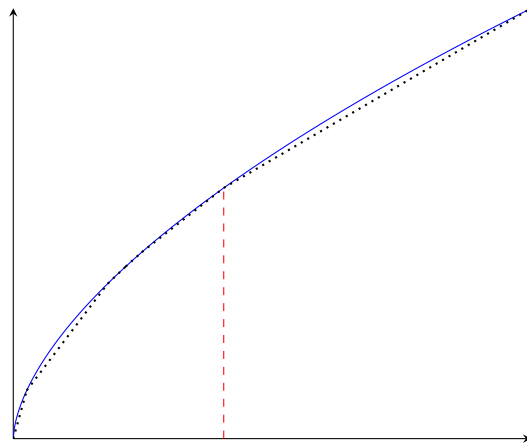
6 A^β Breakpoints

- (1, 4, 2)
 - q : 28.958762
 - $\hat{q}^\beta$: 7.453571
 - q^β : 7.534735
 - Breakpoints: [0, 5.40465749281718, 26.128323085758318, 69.05411764705883]
 - Error: 0.081163827321746
 - Relative error: 0.010771954746023
- (2, 2, 1)
 - q : 30.932783
 - $\hat{q}^\beta$: 7.838833
 - q^β : 7.838833
 - Breakpoints: [0, 1.9849079917037988, 13.879417919656465, 16.512573095835464, 18.890814020083003, 30.93278282491032, 76.14235294117645]
 - Error: 0.000000000000213
 - Relative error: -0.000000000000027
- (2, 4, 1)
 - q : 3.743604
 - $\hat{q}^\beta$: 0.520774
 - q^β : 2.207872
 - Breakpoints: [0, 138.54823529411763]
 - Error: 1.687098048906210
 - Relative error: 0.764128444069464
- (3, 5, 1)
 - q : 71.165974
 - $\hat{q}^\beta$: 12.923003
 - q^β : 12.923003
 - Breakpoints: [0, 30.750913778350093, 44.57376487924321, 71.16597394333036, 181.68117647058827]
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000
- (4, 3, 1)
 - q : 23.670440
 - $\hat{q}^\beta$: 6.676115
 - q^β : 6.676115
 - Breakpoints: [0, 19.921388087224845, 23.670439786681058, 181.14352941176475]
 - Error: 0.000000000000004
 - Relative error: -0.000000000000001

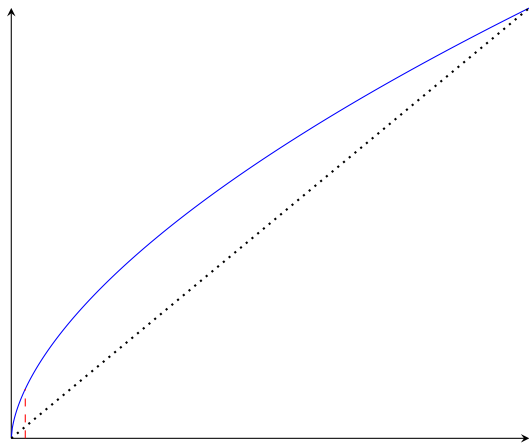
- (5, 1, 1)
 - q : 16.623774
 - $\hat{q}^\beta$: 5.152662
 - q^β : 5.400547
 - Breakpoints: [0, 1.4656333375682882, 13.141872711246277, 89.41176470588236]
 - Error: 0.247884874212676
 - Relative error: 0.045899954259948
- (5, 4, 1)
 - q : 7.000223
 - $\hat{q}^\beta$: 2.518672
 - q^β : 3.214157
 - Breakpoints: [0, 12.878068451030797, 38.682379459353704, 192.3352941176471]
 - Error: 0.695485481818893
 - Relative error: 0.216381911060611
- $cu, 3$
 - q : 16.738920
 - $\hat{q}^\beta$: 5.422960
 - q^β : 5.422961
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 14.376341772690381, 16.738997089244634, 16.738997089244645, 128.44338870924773]
 - Error: 0.000000452192198
 - Relative error: 0.000000083384744
- $cu, 5$
 - q : 20.944873
 - $\hat{q}^\beta$: 6.203641
 - q^β : 6.203641
 - Breakpoints: [0, 6.717165864481, 9.832103793393394, 14.682070969056184, 20.944873037188852, 21.50633939847323, 22.410514202757604, 127.1036185995295]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



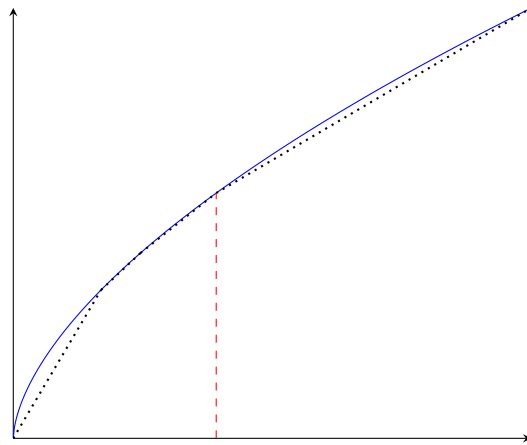
(a) $q^\beta - (1, 4, 2)$



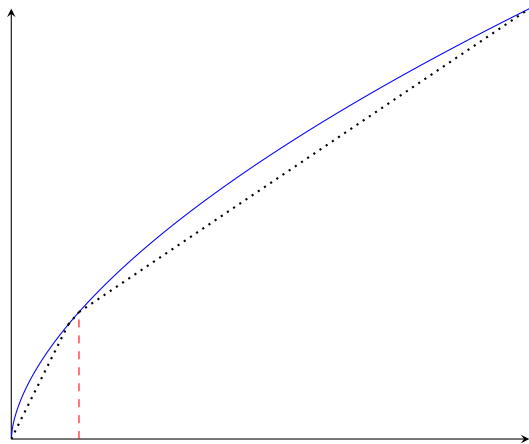
(b) $q^\beta - (2, 2, 1)$



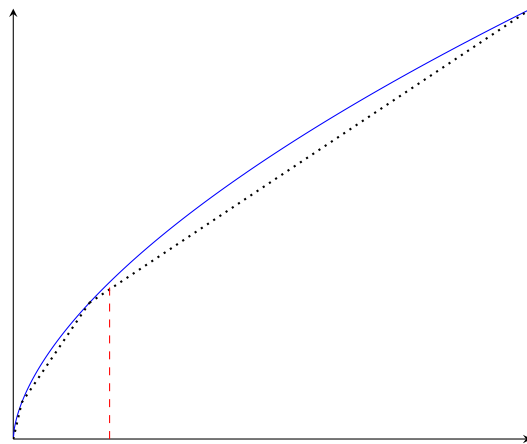
(c) $q^\beta - (2, 4, 1)$



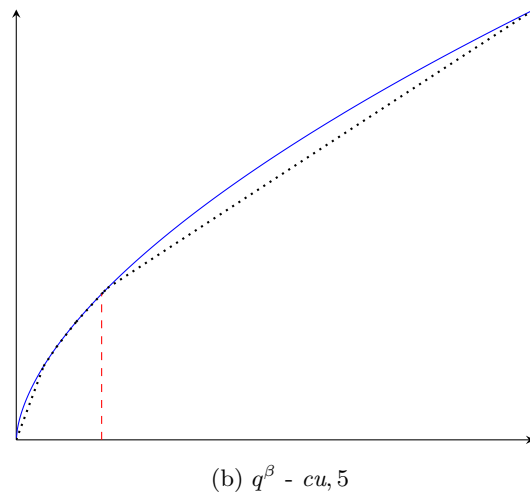
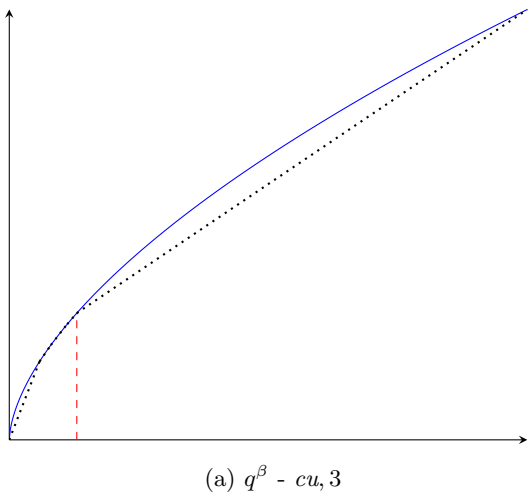
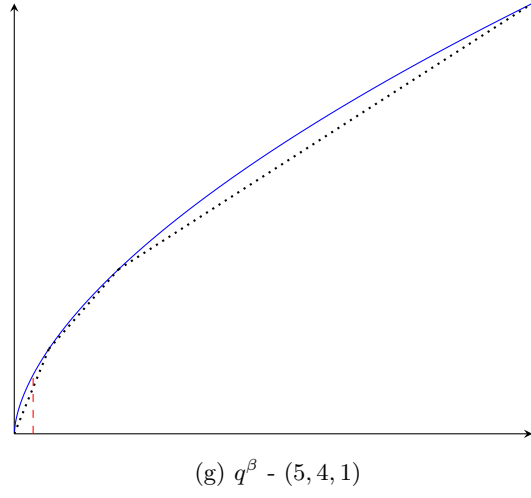
(d) $q^\beta - (3, 5, 1)$



(e) $q^\beta - (4, 3, 1)$



(f) $q^\beta - (5, 1, 1)$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,5,1}$	-0.0	$A_{2,5,1}^\beta$	0.0
$z_{1,1,2}$	-0.0	$z_{4,5,2}$	0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{5,1,2}$	0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,2,2}$	-0.0	$z_{5,2,1}$	0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,2,2}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,3,2}$	-0.0	$z_{5,3,1}$	0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,2}$	-0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,4,2}$	0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,2}$	-0.0	$z_{5,5,1}$	0.0	$A_{3,4,1}^\beta$	0.0
$z_{2,1,1}$	-0.0	$z_{5,5,2}$	-0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,1,2}$	-0.0	$z_{cu,1}$	0.0	$A_{3,5,2}^\beta$	0.0
$z_{2,2,2}$	0.0	$z_{cu,2}$	-0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,1}$	-0.0	$z_{cu,4}$	-0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,3,2}$	-0.0	$z_{hu,1}$	-0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,2}$	-0.0	$z_{hu,2}$	0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,5,1}$	-0.0	$z_{hu,3}$	-0.0	$A_{4,3,2}^\beta$	0.0
$z_{2,5,2}$	-0.0	$z_{hu,4}$	-0.0	$A_{4,4,1}^\beta$	0.0
$z_{3,1,1}$	-0.0	$z_{hu,5}$	-0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	-0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	-0.0	$A_{1,1,2}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	0.0	$A_{1,2,1}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,1}$	-0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,3,2}$	0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,1}$	-0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,4,2}$	-0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{3,5,2}$	-0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,4,2}^\beta$	0.0
$z_{4,1,1}$	-0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,1,2}$	-0.0	$A_{2,1,1}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,1}$	-0.0	$A_{2,1,2}^\beta$	0.0	$A_{cu,1}^\beta$	0.0
$z_{4,2,2}$	-0.0	$A_{2,2,2}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,3,2}$	0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,1}$	-0.0	$A_{2,3,2}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,4,2}$	-0.0	$A_{2,4,2}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^{\beta}$	0.0	$A_{4,3,2}$	0.0	$dt_{1,5,3}$	66.7
$A_{hu,4}^{\beta}$	0.0	$A_{4,4,1}$	0.0	$dt_{2,1,1}$	111.1
$A_{hu,5}^{\beta}$	0.0	$A_{4,4,2}$	0.0	$dt_{2,1,2}$	100.0
$A_{1,1,1}$	0.0	$A_{4,5,1}$	0.0	$dt_{2,1,3}$	100.0
$A_{1,1,2}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,2,3}$	106.1
$A_{1,2,1}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,2,2}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,3,2}$	183.3
$A_{1,3,1}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,3,3}$	183.3
$A_{1,3,2}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,4,3}$	55.6
$A_{1,4,1}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,5,1}$	0.0	$A_{5,4,2}$	0.0	$dt_{2,5,2}$	10.0
$A_{1,5,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{2,5,3}$	111.1
$A_{2,1,1}$	0.0	$A_{5,5,2}$	0.0	$dt_{3,1,1}$	66.7
$A_{2,1,2}$	0.0	$A_{cu,1}$	0.0	$dt_{3,1,2}$	10.0
$A_{2,2,2}$	0.0	$A_{cu,2}$	0.0	$dt_{3,1,3}$	156.75608108108108
$A_{2,3,1}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,3,2}$	0.0	$A_{hu,1}$	0.0	$dt_{3,2,2}$	10.0
$A_{2,4,2}$	0.0	$A_{hu,2}$	0.0	$dt_{3,2,3}$	111.1
$A_{2,5,1}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,5,2}$	0.0	$A_{hu,4}$	0.0	$dt_{3,3,2}$	10.0
$A_{3,1,1}$	0.0	$A_{hu,5}$	0.0	$dt_{3,3,3}$	27.8
$A_{3,1,2}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,1}$	50.0
$A_{3,2,1}$	0.0	$dt_{1,1,2}$	100.0	$dt_{3,4,2}$	117.32775738165911
$A_{3,2,2}$	0.0	$dt_{1,1,3}$	100.0	$dt_{3,4,3}$	144.5
$A_{3,3,1}$	0.0	$dt_{1,2,1}$	17.15006133296988	$dt_{3,5,3}$	133.4
$A_{3,3,2}$	0.0	$dt_{1,2,2}$	10.0	$dt_{4,1,1}$	122.2
$A_{3,4,1}$	0.0	$dt_{1,2,3}$	17.150061332969884	$dt_{4,1,2}$	100.0
$A_{3,4,2}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,3}$	100.0
$A_{3,5,2}$	0.0	$dt_{1,3,2}$	122.2	$dt_{4,2,1}$	122.2
$A_{4,1,1}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,2,2}$	106.1
$A_{4,1,2}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,2,3}$	106.1
$A_{4,2,1}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,3,3}$	183.3
$A_{4,2,2}$	0.0	$dt_{1,5,2}$	10.0	$dt_{4,4,1}$	122.2

Variable	Value	Variable	Value	Variable	Value
$dt_{4,4,2}$	60.57167630057806	$f_{2,1,1}^H$	0.0	$f_{5,5,2}^H$	0.0
$dt_{4,4,3}$	94.5	$f_{2,1,2}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,1}$	122.2	$f_{2,2,2}^H$	0.0	$f_{1,1,2}^C$	0.0
$dt_{4,5,2}$	111.1	$f_{2,3,1}^H$	0.0	$f_{1,2,1}^C$	0.0
$dt_{4,5,3}$	55.6	$f_{2,3,2}^H$	0.0	$f_{1,2,2}^C$	0.0
$dt_{5,1,3}$	138.9	$f_{2,4,2}^H$	0.0	$f_{1,3,1}^C$	0.0
$dt_{5,2,1}$	74.66004781348396	$f_{2,5,1}^H$	0.0	$f_{1,3,2}^C$	0.0
$dt_{5,2,2}$	10.0	$f_{2,5,2}^H$	10.6	$f_{1,4,1}^C$	0.0
$dt_{5,2,3}$	21.35698491649854	$f_{3,1,1}^H$	0.0	$f_{1,5,1}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,1,2}^H$	0.0	$f_{1,5,2}^C$	0.0
$dt_{5,3,2}$	10.0	$f_{3,2,1}^H$	5.2140906269397895	$f_{2,1,1}^C$	0.0
$dt_{5,3,3}$	31.000725395828823	$f_{3,2,2}^H$	0.0	$f_{2,1,2}^C$	0.0
$dt_{5,4,3}$	10.0	$f_{3,3,1}^H$	0.0	$f_{2,2,2}^C$	6.1
$dt_{5,5,1}$	105.6	$f_{3,3,2}^H$	0.0	$f_{2,3,1}^C$	0.0
$dt_{5,5,2}$	10.0	$f_{3,4,1}^H$	0.0	$f_{2,3,2}^C$	8.4
$dt_{5,5,3}$	10.0	$f_{3,4,2}^H$	0.0	$f_{2,4,2}^C$	0.0
$dt_{cu,1}$	53.3	$f_{3,5,2}^H$	14.8	$f_{2,5,1}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,1}^H$	0.0	$f_{2,5,2}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,1,2}^H$	0.0	$f_{3,1,1}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,1}^H$	0.0	$f_{3,1,2}^C$	7.6
$dt_{hu,2}$	18.3	$f_{4,2,2}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,3,2}^H$	12.6	$f_{3,2,2}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,1}^H$	0.0	$f_{3,3,1}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,4,2}^H$	0.0	$f_{3,3,2}^C$	0.0
$f_{1,1,1}^H$	8.8	$f_{4,5,1}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,1,2}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,4,2}^C$	4.8566217507962115
$f_{1,2,1}^H$	0.0	$f_{5,1,2}^H$	0.0	$f_{3,5,2}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,2,1}^H$	0.0	$f_{4,1,1}^C$	0.0
$f_{1,3,1}^H$	0.0	$f_{5,2,2}^H$	7.231133337143549	$f_{4,1,2}^C$	0.0
$f_{1,3,2}^H$	0.0	$f_{5,3,1}^H$	0.0	$f_{4,2,1}^C$	0.0
$f_{1,4,1}^H$	0.0	$f_{5,3,2}^H$	10.468866662856449	$f_{4,2,2}^C$	0.0
$f_{1,5,1}^H$	0.0	$f_{5,4,2}^H$	0.0	$f_{4,3,2}^C$	0.0
$f_{1,5,2}^H$	0.0	$f_{5,5,1}^H$	8.116804201050268	$f_{4,4,1}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{4,4,2}^C$	0.0	$RecLMTD_{3,3,2}$	0.02645080077831024	$q_{1,3,1}$	0.0
$f_{4,5,1}^C$	0.0	$RecLMTD_{3,4,1}$	0.011870193451542476	$q_{1,3,2}$	0.0
$f_{4,5,2}^C$	13.9	$RecLMTD_{3,4,2}$	0.0075710837422526284	$q_{1,4,1}$	0.0
$f_{5,1,2}^C$	0.0	$RecLMTD_{3,5,2}$		$q_{1,5,1}$	0.0
$f_{5,2,1}^C$	0.0	$RecLMTD_{4,1,1}$	0.020350921041481694	$q_{1,5,2}$	0.0
$f_{5,2,2}^C$	0.0	$RecLMTD_{4,1,2}$	0.009908927464973554	$q_{2,1,1}$	0.0
$f_{5,3,1}^C$	0.0	$RecLMTD_{4,2,1}$		$q_{2,1,2}$	0.0
$f_{5,3,2}^C$	0.0	$RecLMTD_{4,2,2}$	0.008674619618294832	$q_{2,2,2}$	0.0
$f_{5,4,2}^C$	0.0	$RecLMTD_{4,3,2}$		$q_{2,3,1}$	0.0
$f_{5,5,1}^C$	0.0	$RecLMTD_{4,4,1}$	0.010870664658609884	$q_{2,3,2}$	0.0
$f_{5,5,2}^C$	0.0	$RecLMTD_{4,4,2}$	0.012271026275456285	$q_{2,4,2}$	0.0
$RecLMTD_{1,1,1}$	0.1	$RecLMTD_{4,5,1}$	0.008069446003894656	$q_{2,5,1}$	0.0
$RecLMTD_{1,1,2}$	0.1	$RecLMTD_{4,5,2}$	0.01184615087992687	$q_{2,5,2}$	0.0
$RecLMTD_{1,2,1}$	0.1	$RecLMTD_{5,1,2}$		$q_{3,1,1}$	0.0
$RecLMTD_{1,2,2}$	0.0642496933351506	$RecLMTD_{5,2,1}$		$q_{3,1,2}$	0.0
$RecLMTD_{1,3,1}$	0.1	$RecLMTD_{5,2,2}$	0.043215075417507326	$q_{3,2,1}$	0.0
$RecLMTD_{1,3,2}$	0.008183306055646482	$RecLMTD_{5,3,1}$		$q_{3,2,2}$	0.0
$RecLMTD_{1,4,1}$	0.1	$RecLMTD_{5,3,2}$	0.028888218232147517	$q_{3,3,1}$	0.0
$RecLMTD_{1,5,1}$	0.1	$RecLMTD_{5,4,2}$		$q_{3,3,2}$	0.0
$RecLMTD_{1,5,2}$	0.033467722397311	$RecLMTD_{5,5,1}$		$q_{3,4,1}$	0.0
$RecLMTD_{2,1,1}$	0.00943853840668675	$RecLMTD_{5,5,2}$		$q_{3,4,2}$	0.0
$RecLMTD_{2,1,2}$	0.1	$RecLMTD_{cu,1}$	0.01610890859262676	$q_{3,5,2}$	0.0
$RecLMTD_{2,2,2}$	0.016168655950305852	$RecLMTD_{cu,2}$	0.009423141115769086	$q_{4,1,1}$	0.0
$RecLMTD_{2,3,1}$	0.006171009673674765	$RecLMTD_{cu,4}$	0.008535688158765427	$q_{4,1,2}$	0.0
$RecLMTD_{2,3,2}$	0.005360924403785095	$RecLMTD_{hu,1}$	0.012387051800747542	$q_{4,2,1}$	0.0
$RecLMTD_{2,4,2}$	0.1	$RecLMTD_{hu,2}$	0.045618305265862594	$q_{4,2,2}$	0.0
$RecLMTD_{2,5,1}$	0.022648232592241375	$RecLMTD_{hu,3}$	0.04305097282506776	$q_{4,3,2}$	0.0
$RecLMTD_{2,5,2}$	0.022648232592241375	$RecLMTD_{hu,4}$	0.015434474530129737	$q_{4,4,1}$	0.0
$RecLMTD_{3,1,1}$	0.025585643678262784	$RecLMTD_{hu,5}$	0.024141568940966462	$q_{4,4,2}$	0.0
$RecLMTD_{3,1,2}$	0.01871414246749941	$q_{1,1,1}$		$q_{4,5,1}$	0.0
$RecLMTD_{3,2,1}$	0.1	$q_{1,1,2}$		$q_{4,5,2}$	0.0
$RecLMTD_{3,2,2}$	0.023816474813566153	$q_{1,2,1}$		$q_{5,1,2}$	0.0
$RecLMTD_{3,3,1}$	0.1	$q_{1,2,2}$		$q_{5,2,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{5,2,2}$	0.0	$t_{3,2,1}^H$	226.7	$t_{2,1,1}^C$	129.7960526315789
$q_{5,3,1}$	0.0	$t_{3,2,2}^H$	65.6	$t_{2,1,2}^C$	129.79605263157896
$q_{5,3,2}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,2,2}^C$	115.6
$q_{5,4,2}$	0.0	$t_{3,3,2}^H$	65.6	$t_{2,3,1}^C$	37.8
$q_{5,5,1}$	0.0	$t_{3,4,1}^H$	116.23716216216218	$t_{2,3,2}^C$	37.8
$q_{5,5,2}$	0.0	$t_{3,4,2}^H$	65.6	$t_{2,4,2}^C$	112.86184971098265
$q_{cu,1}$	0.0	$t_{3,5,2}^H$	122.35608108108107	$t_{2,5,1}^C$	93.3
$q_{cu,2}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,5,2}^C$	131.46187050359833
$q_{cu,4}$	0.0	$t_{4,1,2}^H$	148.9	$t_{3,1,1}^C$	60.0
$q_{hu,1}$	0.0	$t_{4,2,1}^H$	237.58333333333334	$t_{3,1,2}^C$	60.0
$q_{hu,2}$	0.0	$t_{4,2,2}^H$	148.9	$t_{3,2,1}^C$	115.6
$q_{hu,3}$	0.0	$t_{4,3,2}^H$	148.9	$t_{3,2,2}^C$	115.6
$q_{hu,4}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,3,1}^C$	111.5
$q_{hu,5}$	0.0	$t_{4,4,2}^H$	148.9	$t_{3,3,2}^C$	37.8
$t_{1,1,1}^H$	160.0	$t_{4,5,1}^H$	148.9	$t_{3,4,1}^C$	82.2
$t_{1,1,2}^H$	93.3	$t_{4,5,2}^H$	148.9	$t_{3,4,2}^C$	82.2
$t_{1,2,1}^H$	160.0	$t_{5,1,2}^H$	65.6	$t_{3,5,2}^C$	166.23812949640183
$t_{1,2,2}^H$	93.3	$t_{5,2,1}^H$	198.9	$t_{4,1,1}^C$	160.0
$t_{1,3,1}^H$	160.0	$t_{5,2,2}^H$	102.51642558256174	$t_{4,1,2}^C$	60.0
$t_{1,3,2}^H$	93.3	$t_{5,3,1}^H$	198.9	$t_{4,2,1}^C$	184.83114754098358
$t_{1,4,1}^H$	160.0	$t_{5,3,2}^H$	89.81182300500888	$t_{4,2,2}^C$	115.6
$t_{1,5,1}^H$	160.0	$t_{5,4,2}^H$	100.01169124992158	$t_{4,3,2}^C$	37.8
$t_{1,5,2}^H$	93.3	$t_{5,5,1}^H$	198.9	$t_{4,4,1}^C$	82.2
$t_{2,1,1}^H$	137.8	$t_{5,5,2}^H$	65.6	$t_{4,4,2}^C$	82.2
$t_{2,1,2}^H$	137.8	$t_{1,1,1}^C$	90.20394736842101	$t_{4,5,1}^C$	204.4
$t_{2,2,2}^H$	137.8	$t_{1,1,2}^C$	144.10854085872572	$t_{4,5,2}^C$	93.3
$t_{2,3,1}^H$	159.0169811320755	$t_{1,2,1}^C$	115.6	$t_{5,1,2}^C$	60.0
$t_{2,3,2}^H$	137.8	$t_{1,2,2}^C$	115.6	$t_{5,2,1}^C$	115.6
$t_{2,4,2}^H$	137.8	$t_{1,3,1}^C$	111.5	$t_{5,2,2}^C$	115.6
$t_{2,5,1}^H$	198.85754716981074	$t_{1,3,2}^C$	107.67619047619047	$t_{5,3,1}^C$	111.5
$t_{2,5,2}^H$	137.8	$t_{1,4,1}^C$	102.85953757225435	$t_{5,3,2}^C$	37.8
$t_{3,1,1}^H$	65.6	$t_{1,5,1}^C$	148.85	$t_{5,4,2}^C$	146.03815028901732
$t_{3,1,2}^H$	65.6	$t_{1,5,2}^C$	93.3	$t_{5,5,1}^C$	148.85

Variable	Value	Variable	Value	Variable	Value
$t_{5,5,2}^C$	93.3	$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	1408.0	$b_{4,5,1}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,1,2}^{H,in}$	0.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0
$b_{1,2,1}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	0.0	$b_{3,5,2}^{H,out}$	1810.87
$b_{1,2,2}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	0.0	$b_{4,1,1}^{H,out}$	0.0
$b_{1,3,1}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	1127.7830797279596	$b_{4,1,2}^{H,out}$	0.0
$b_{1,3,2}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	0.0	$b_{4,2,1}^{H,out}$	0.0
$b_{1,4,1}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	1115.30692027204	$b_{4,2,2}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	0.0	$b_{5,4,2}^{H,in}$	0.0	$b_{4,3,2}^{H,out}$	1876.1399999999994
$b_{1,5,2}^{H,in}$	0.0	$b_{5,5,1}^{H,in}$	1614.4323555888984	$b_{4,4,1}^{H,out}$	0.0
$b_{2,1,1}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	0.0	$b_{4,4,2}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	1408.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	0.0	$b_{1,1,2}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	0.0	$b_{1,2,1}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,4,2}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	1127.7830797279596
$b_{2,5,1}^{H,in}$	0.0	$b_{1,3,2}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	0.0
$b_{2,5,2}^{H,in}$	1460.68	$b_{1,4,1}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	1115.30692027204
$b_{3,1,1}^{H,in}$	0.0	$b_{1,5,1}^{H,out}$	0.0	$b_{5,4,2}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	0.0	$b_{1,5,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	1614.4323555888986
$b_{3,2,1}^{H,in}$	1182.0343451272502	$b_{2,1,1}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	0.0
$b_{3,2,2}^{H,in}$	0.0	$b_{2,1,2}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,3,1}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	0.0	$b_{1,1,2}^{C,in}$	0.0
$b_{3,3,2}^{H,in}$	0.0	$b_{2,3,1}^{H,out}$	0.0	$b_{1,2,1}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,4,2}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0
$b_{3,5,2}^{H,in}$	1810.87	$b_{2,5,1}^{H,out}$	0.0	$b_{1,3,2}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	0.0	$b_{2,5,2}^{H,out}$	1460.68	$b_{1,4,1}^{C,in}$	0.0
$b_{4,1,2}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	0.0	$b_{1,5,1}^{C,in}$	0.0
$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	0.0	$b_{1,5,2}^{C,in}$	0.0
$b_{4,2,2}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	1182.0343451272502	$b_{2,1,1}^{C,in}$	0.0
$b_{4,3,2}^{H,in}$	1876.1399999999994	$b_{3,2,2}^{H,out}$	0.0	$b_{2,1,2}^{C,in}$	0.0
$b_{4,4,1}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0	$b_{2,2,2}^{C,in}$	705.16

Variable	Value	Variable	Value	Variable	Value
$b_{2,3,1}^{C,in}$	0.0	$b_{1,2,1}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,3,2}^{C,in}$	317.52	$b_{1,2,2}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{2,4,2}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,3,1}^{C,out}$	0.0
$b_{2,5,2}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	0.0	$b_{1,5,1}^{C,out}$	0.0	$b_{5,4,2}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	456.0	$b_{1,5,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{2,1,1}^{C,out}$	0.0	$b_{5,5,2}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{2,1,2}^{C,out}$	0.0		
$b_{3,3,1}^{C,in}$	0.0	$b_{2,2,2}^{C,out}$	705.16		
$b_{3,3,2}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	0.0		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	317.52		
$b_{3,4,2}^{C,in}$	399.2143079154486	$b_{2,4,2}^{C,out}$	0.0		
$b_{3,5,2}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,5,2}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	0.0	$b_{3,1,1}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	456.0		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,3,2}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,4,1}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	1296.8700000000001	$b_{3,4,2}^{C,out}$	399.2143079154486		
$b_{5,1,2}^{C,in}$	0.0	$b_{3,5,2}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		
$b_{5,2,2}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	0.0		
$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0		
$b_{5,3,2}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0		
$b_{5,4,2}^{C,in}$	0.0	$b_{4,3,2}^{C,out}$	0.0		
$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	0.0		
$b_{5,5,2}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0		
$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,1}^{C,out}$	0.0		
$b_{1,1,2}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	1296.8700000000001		